

LAGRANGIAN SIMULATION OF FIRE PLUMES

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ABSTRACT

This paper describes recent progress in the formulation of a Lagrangian computational approach to simulate large fire plumes, and results of the application of this approach to a representative case. The approach is based on the vortex-element method in which the unsteady Navier-Stokes equations are solved in their vorticity-transport form using a set of moving computational elements carrying time-dependent vorticity. The scheme is grid-free and the computational elements move along particle trajectories thus endowing the scheme with an adaptive property. Vorticity is generated due to the buoyancy-induced torque according to the evolving density and pressure gradients. The energy and species transport equations are solved using the transport-element method in which, to maintain compatibility with the vortex method, the gradients of the scalars are evolved by convection and diffusion. The fuel oxidation reaction is assumed to be infinitely fast and Schvab-Zeldovich variables are used to reduce the combustion formulation to a single transport equation. The approach is applied to compute the evolution of a fire plume, given the initial fuel velocity at the pool surface. Results show that the plume dynamics are dominated by the shedding of large vortical structures which originate in the vertical vorticity layer that forms above the pool. While the formation rate of vorticity is buoyancy dominated, the instability resembles that of the Kelvin-Helmholtz type. The structures appear to control the entrainment process, and burning occurs on their outer edges. The diffusion flame, whose thickness, except for zones of intense strain close to the large eddies, increases gradually from the pool edge upwards. Within the large eddies, the flame executes almost a full turn around the large structures, and produces zones of high fuel concentration inside the structures. The fuel is 'pinched' off the rising plume in large puffs as the entrained air reaches the plume centerline, causing the flame height to oscillate at the same frequency as the shed structures. Predictions of the puffing frequency, averaging flame height and oscillations, and motion of the large structures are compared with experimental data.

1. INTRODUCTION

Large fires occur when a large supply of fuel which may be in the form of oil spill, an oil well, a dense forest, etc., is ignited. Besides their danger to human life and material,

massive fires represent an environmental hazard, especially when they are induced for the purpose of cleaning up fuel spills or disposing of unwanted debris. In such cases, the composition of the combustion products, the soot concentration and the amount of unburned or partially burned fuel injected into the atmosphere is of major concern. Since fires are naturally ventilated combustion phenomena, the quality of combustion in these processes depends strongly on the flow field they establish, and the amount of air they entrain into the flame zone. Thus, the problem of determining the burning rate and the fire impact on the environment becomes strongly tied to determining the flow and combustion field in the neighborhood of the fuel pool.

Experimental investigation of laboratory fires, which maintain the characteristics of massive fires, has been conducted over a range of parameters [1,2] for the purpose of investigating their dynamics and the factors determining the composition of their products. Experimental [1,2] and numerical studies [3-5] have shown that a large fire possesses strong dynamics which can be characterized by the shedding of large burning structures near the pool. The formation of these structures is believed to impact the entrainment and hence the burning rate and products composition strongly. These findings suggest that numerical simulations of fires must rely on approaches capable of capturing these structures. The effort described in this paper, whose objective is to develop a fire plume computational model that incorporates the major physical interactions involved: buoyancy, combustion, and radiation, utilizes an approach which has shown to be capable of capturing large unsteady flow structures.

The first step in this endeavor was to compute the flow in an isothermal plume using a vorticity based formulation [6,7]. Results of this formulation showed good agreement with experimental results on nonreacting plumes. In the next step, combustion was introduced using an infinite rate formulation of the chemical reaction [3,4]. The dynamic effects of combustion, including volumetric expansion and vorticity generation were also included in the formulation. To account for the effect of radiation, including heating the fuel pool and reducing the heat added to the plume, we reduced the enthalpy of reaction by 35% of its adiabatic value. Fundamental simulation of radiation is currently pursued and will be introduced into the model

in the future. In this paper we report on this infinite chemistry results of fire plume simulations, and focus on the entrainment field created by the formation of the large scale structures. As shown in Sections II and III, since our last publication in the area, we have improved the numerical simulation significantly by introducing the proper core function for the computational elements

The paper is organized as follow. The vortex-element method, the basis for computing the buoyant fluid mechanics, is discussed in Section II. The transport-element method, which is used to compute the transport of nonreacting and reacting scalars, is presented in Section III. An example is presented and analyzed in Section IV. Conclusions and suggestions for future work are summarized in Section V.

II THE VORTEX ELEMENT METHOD

The fire plume flowfield is largely inviscid, turbulent and unsteady. Except for the ground, it is unbounded. The shear, or high velocity gradient zone exists within a narrow area where a strong density gradient is encountered. The approach we choose to simulate this flowfield is the vortex method for a low Mach number compressible flow. In an axisymmetric domain (r, z) , the governing the continuity and momentum equations, written in dimensionless form, are:

$$\frac{D\rho}{Dt} + \rho \nabla \cdot \vec{u} = 0 \quad (1)$$

$$\frac{Du}{Dt} = -\frac{1}{\rho} \frac{\partial p}{\partial r} + \frac{1}{Re} \left[\nabla^2 u - \frac{u}{r} \right] \quad (2)$$

$$\frac{Dw}{Dt} = -\frac{1}{\rho} \frac{\partial p}{\partial z} + \frac{1}{Re} \nabla^2 w \quad (3)$$

where the velocity $\vec{u} = (u, w)$, u is radial r -component and w is in the axial z -component, nondimensionalized with respect to $U^* = \sqrt{gD}$, g and D are gravitational acceleration and pool diameter, respectively. The Reynold number is $Re = DU^*/\nu$, ν being the kinematic viscosity. The density, ρ , is nondimensionalized with respect to the ambient density ρ_a . The difference between the local pressure and the atmospheric pressure is p , nondimensionalized with respect to $\rho_a g D$.

The essence of the vortex-element method is to represent the vorticity field using a number of computational vortex elements, update the vorticity of the elements as they move with the flow according to the torque induced by the cross product of the pressure and density gradient, and calculate the vorticity-induced velocity of each element using the Biot-Savart law by summing the contributions of all the elements [8,9]. One of the main advantages of the vortex-element method is that one avoids the direct discretization of the spatial velocity gradients and hence the introduction of numerical dif-

fusion. The method is also an optimal adaptive grid scheme since the computational elements exist only in regions of non-zero vorticity. Therefore, one is able to simulate high Reynolds number flows using a finite number of computational elements [10].

The equation governing the time evolution of the circulation of an element is obtained by integrating the vorticity equation over the area, δA , the latter obtained by taking the curl of the momentum equations (2,3) and using the continuity equation (1):

$$\frac{D\Gamma}{Dt} = \delta A \left[\frac{\partial}{\partial r} \left(\frac{1}{\rho} \right) \frac{\partial p}{\partial z} - \frac{\partial}{\partial z} \left(\frac{1}{\rho} \right) \frac{\partial p}{\partial r} - \frac{\partial}{\partial r} \left(\frac{1}{\rho} \right) \right] + \frac{\delta A}{Re} \left(\nabla^2 \omega - \frac{\omega}{r} \right) \quad (4)$$

The circulation equation is solved in two steps: advection with circulation update, and (2) viscous diffusion. The radial and axial gradients of the pressure are obtained from the momentum equations (2,3) without the diffusion terms. The spacial gradients of the inverse of the density are obtained using the transport-element method described in Section III.

The vorticity field is modeled as a collection of vortex elements distributed along a number of layers initially placed at the interface between the fuel and oxidizer, as shown in figure 1. These elements are transported in time according to:

$$\frac{d\vec{\chi}}{dt} = \vec{u}, \text{ where } \vec{\chi} \text{ is the location of a computational element}$$

and the velocity $\vec{u}$ is obtained from the Helmholtz decomposition as discussed next. Once the elements locations are found, the circulation is updated using equation (4).

The connectivity between the elements describing a layer, and the condition that the interdistance between neighboring elements must not exceed the core radius, are maintained by inserting elements whenever two neighboring elements move further apart due to stretching. That core overlap must always be maintained by choosing the core radius to be larger than the interdistance between neighboring elements guarantees the smooth distribution of the computed vorticity field. Insertion of an element is performed such that properties are conserved and that the distance separating neighboring elements is of the order of the initial spatial discretization [8,10]. For an element at a certain location and of a length δl , the physical area δA and the element thickness δn are updated via the conservation of mass. In the case of buoyancy-driven plume, the initial circulation for all the elements is zero value since inertia effects are negligible. The initial velocity field is the potential field induced by the outflow at the pool surface.

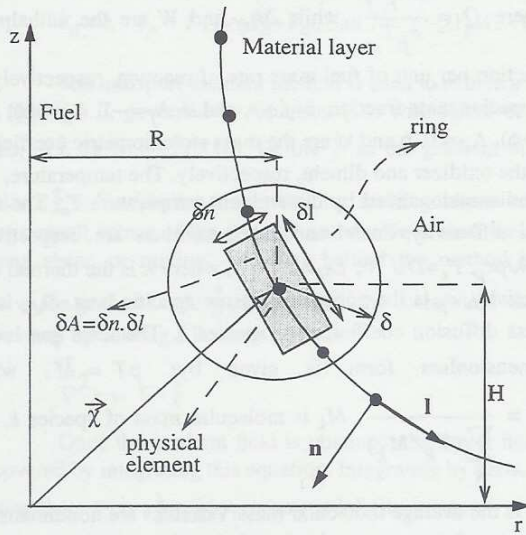


Figure 1. Vortex element in an axisymmetric domain.

The velocity is expressed in terms of its components using the Helmholtz decomposition:

$$\vec{u} = \vec{u}_p + \vec{u}_\omega + \vec{u}_e \quad (5)$$

where $\vec{u}_p$, $\vec{u}_\omega$, and $\vec{u}_e$ are the potential, vortical, and volumetric expansion components of the velocity, respectively. The potential field is obtained from $\vec{u}_p = \nabla\Phi$, where the velocity potential, Φ , satisfies the Laplace equation: $\nabla^2\Phi = 0$ with B.C. $\frac{\partial\Phi}{\partial n} = -(\vec{u}_\omega + \vec{u}_e) \cdot \vec{n}$ at the solid boundaries. The component of the potential flow induced by the outflow at the pool is obtained using the panel method, once and for all, on an Eulerian grid covering the entire computational domain. Interpolation is used to evaluate this component at an arbitrary position. The potential component due to vorticity is calculated by the method of images, where the image is symmetrical to the element with respect to the solid boundary, the ground in our case.

The velocity induced by the vorticity is obtained by taking the curl of the stream function, Ψ , which satisfies the Poisson equation, $\nabla^2\Psi = -\vec{\omega}$ where $\vec{\omega} = (0, 0, \omega)$. The velocity induced by volumetric expansion is obtained from its potential, Φ_e which satisfies the Poisson equation with the Lagrangian time variation of the density as the source: $\nabla^2\Phi_e = -\frac{1}{\rho}\frac{D\rho}{Dt}$.

In axisymmetric coordinates, the vorticity is discretized into a number of vortex rings, each with a circulation Γ_i [11],

$$\omega(\vec{\chi}, t) = \sum_{j=1}^N \frac{\Gamma_j}{\pi\delta_j^2} f_\delta \left(\frac{|\vec{\chi} - \vec{\chi}_j|}{\delta_j} \right) \quad (6)$$

and $\Gamma_i = \omega_i(\vec{\chi}_i, t)A_i$. The total vorticity associated with a vortex ring is defined by its circulation Γ_i , while its distribution is given by f_δ , the core function, and δ_i is the core radius. Similar distribution is used to describe $D\rho/Dt$, computed using the rate of change of the element density due to combustion along the reaction surface.

The streamfunction in axisymmetric coordinates, at an arbitrary point $\vec{\chi}(r, z, \alpha)$, for is given by:

$$\Psi(\vec{\chi}, t) = \sum_{j=1}^N \frac{\Gamma_j}{\pi\delta_j^2} \int_V G(\vec{\chi} - \vec{\chi}') f_\delta \left(\frac{|\vec{\chi} - \vec{\chi}_j|}{\delta_j} \right) d\vec{\chi}' \quad (7)$$

$G(\vec{\chi} - \vec{\chi}')$ is the Green's function $G(\vec{\chi} - \vec{\chi}') = \frac{1}{4\pi|\vec{\chi} - \vec{\chi}'|}$, where V is the volume of the vortex ring. Consider a point P with position vector $\vec{\chi}(r, z, 0)$, and a vortex ring j located at $\vec{\chi}_j(0, H)$. Using the complete elliptic integrals of the first and second kind of modulus k , $K(k)$, and $E(k)$, where $k^2 = \frac{4rp}{(z-\zeta)^2 + (r+\rho)^2}$, where $\rho = R + s \cos\theta$, $\zeta = H + s \sin\theta$, and representing the core in terms of the local (s, θ) coordinates, equation (7) is written as:

$$\Psi = \sum_{j=1}^N \frac{\Gamma_j}{\pi R_j} \int_0^{2\pi} \int_0^{\delta_j} s(R_j + s \cos\theta) f_{\delta_j}(s, \theta) G'(s, \theta) ds d\theta \quad (8)$$

$$\text{where } G'(s, \theta) = \frac{\sqrt{r(R_j + s \cos\theta)}}{2\pi} \left[\left(\frac{2}{k} - k \right) K(k) - \frac{2}{k} E(k) \right].$$

In the case of a thin ring, $R \gg \delta$, the vortical velocity induced at a point away from the core center, can be obtained by assuming that the circulation of the vortex ring is concentrated along its centerline, i.e. circular vortex filament. The stream function and the corresponding vortical velocity field can be derived easily from equation (8).

For points close and within the core, incorporating a smoothening core function is essential to obtain non singular and accurate solutions. A Gaussian core function of the form

$$f_\delta = e^{-(s/\delta)^2} \text{ has been used. In Cartesian coordinates this}$$

core function is second-order accurate, smooth, and satisfies the vorticity equation for an inertia-free flow induced by a thin vortex ring, which is compatible with the core expansion algorithm used to model viscous diffusion [4,5]. The vortical and volumetric expansion induced velocities (and the scalar field, as will be shown next) are obtained by integrating equation (8) numerically for different values of $(R/\delta, r/\delta, \xi/\delta)$. To achieve maximum computational efficiency, this requires constructing tables for the dimensionless velocity (and scalar components) at values of (r, ξ) as R/δ . Three-dimensional tables, covering three independent variables $r/\delta, \xi/\delta$, and R/δ are time consuming for data access. The alternative is to represent the variation with R/δ of a certain parameter at location $(r/\delta, \xi/\delta)$ by a polynomial of n th degree. The degree of the polynomial is selected so that the error induced by fitting is negligible. (See Ref [4] for more detail).

Substituting the element vorticity into the diffusion equation yields the following time evolution equation for δ^2 ,

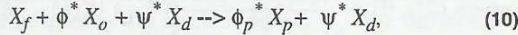
$\frac{d(\delta^2)}{dt} = 4\nu$. Thus, viscous diffusion is modeled by core expansion according to:

$$\delta^2 = \delta_o^2 + 4\nu t \quad (9)$$

III. THE TRANSPORT ELEMENT METHOD

Isothermal and reactive plumes are characterized by sharp density gradients which, as can be seen from equation (4) determine the evolution of the vorticity field in these flows through the gravity and baroclinic vorticity generation terms. For the purpose of evaluating the evolution of the density field in a way compatible with the vortex method, i.e. in a Lagrangian and grid-free way, we have developed the transport-element method and used it to compute 2-D and 3-D non-reacting and reacting flows in Cartesian coordinates. In this section, we describe the transport-element method in an axisymmetric domain [12,13].

The chemical reaction is assumed to be single-step,



where X is species, f stands for fuel, o for oxidizer, p for product, and d for diluent, and $\phi^* (=n_o/n_f)$ is the molar stoichiometric coefficient while $\psi^* (=n_d/n_f)$ is the diluent/oxidant molar fraction. The energy and species conservation equations are expressed in dimensionless form as follows:

$$\frac{DT}{Dt} = \frac{1}{Pe} \nabla^2 T + Q \frac{W}{\rho} \quad (11)$$

$$\frac{DY_k}{Dt} = \frac{1}{Le_k Pe} \nabla^2 Y_k + \Lambda_k \frac{W}{\rho} \quad (12)$$

where $Q = \frac{\Delta h_{r,f}}{c_p T_o}$, while $\Delta h_{r,f}$ and W are the enthalpy of

reaction per unit of fuel mass rate of reaction, respectively, Y_k is species mass fraction, $k=f,o,p$, and d . $\Lambda_f = -1$, $\Lambda_o = -\phi$, $\Lambda_p = (1+\phi)$, $\Lambda_d = 0$, ϕ and ψ are the mass stoichiometric coefficients of the oxidizer and diluent, respectively. The temperature, T , is nondimensionalized by the ambient temperature T_o . The thermal diffusivity, Peclet and Lewis numbers are, respectively: $\alpha = \lambda / \rho c_p$, $Pe = DU^* / \alpha$, $Le_k = \alpha / D_{d,k}$, where λ is the thermal conductivity, c_p is the constant pressure specific heat, $D_{d,k}$ is the mass diffusion coefficient of species k . The ideal gas law in dimensionless form is given by: $\rho T = \bar{M}$, where

$$\bar{M} = \frac{1}{\sum_k (Y_k / M_k)}, \quad M_k \text{ is molecular mass of species } k, \text{ and}$$

$\bar{M}$ is the average molecular mass. Variables are nondimensionalized with respect to values on the oxidizer side [3].

The temperature and species concentrations equations are complicated by the presence of source terms. These "reactive scalar" equations can be reduced to conserved scalar using the Schvab-Zeldovich variables [10,11], which are constructed from combinations of the primitive reactive scalars such that their transport equations do not contain source terms [12,14,15]. We define the following S-Z variables:

$$\lambda = Y_o - \phi Y_f, \quad \gamma = T - \frac{Q}{1+\phi} Y_p, \quad \text{and} \quad \delta = Y_d \quad (13)$$

In the case $Le = 1$ and $D_{d,k} = D_d$, the transport equations of such variables, obtained from algebraic manipulations of equations (16 and 17), have the following general form:

$$\frac{d\beta}{dt} = \frac{1}{Pe} \nabla^2 \beta \quad (14)$$

where β is a S-Z variable. Now, define a normalized S-Z variable; the mixture fraction:

$$\zeta = \frac{\beta - \beta_o}{\beta_f - \beta_o} \quad (15)$$

for $\beta = \lambda, \gamma$, and δ , where the subscripts o and f refer to the initial condition on the fuel side and oxidizer side of the plume, respectively. The normalized S-Z scalar is governed by equation (19) with the same boundary conditions: $\zeta = 1$ on the fuel side, and 0 on the oxidizer side.

In the case of infinite-rate chemistry, the fuel and oxidizer do not co-exist and the reaction occurs across a reaction interface where the products are produced. At this reaction interface, $Y_f = 0$ and $Y_o = 0$, leading to $\lambda = 0$. Thus:

at $\lambda = 0$: the reaction interface, and $Y_f = Y_o = 0$,

at $\lambda > 0$, the oxidizer side, and $Y_f = 0$, $Y_o = \lambda$,

at $\lambda < 0$, the fuel side, and $Y_f = -\lambda\phi$, $Y_o = 0$,
 $Y_d = \delta$, $Y_p = 1 - (Y_f + Y_o + Y_d)$, and $T = \gamma + QY_p / (1 + \phi)$.

The transport-element method is used to evolve a scalar field $s = \zeta$, governed by equation (14) with initial conditions: $s(\vec{x}, 0) = s_o(\vec{x})$ [10]. Define $\vec{g}$ as the gradient of the scalar s , $\vec{g} = \nabla s$. Similar to the vorticity, scalar gradients exist over a small subset of the domain, and evolve around areas of strong shear, or mixing. The idea behind the method is to evolve the gradient field $\vec{g}$, from which the scalar field can be recovered by solving a Poisson equation:

$$\nabla^2 s = \nabla \cdot \vec{g} \quad (16)$$

Once the gradient field is obtained, the scalar field is recovered by integrating this equation. Integrating by parts, and noting that $\nabla(G \cdot \vec{g})$ goes to zero at infinity, we get:

$$s(\vec{x}, t) = \int_V \vec{g}(\vec{x}', t) \cdot \nabla G(\vec{x} - \vec{x}') d\vec{x}' + s_p(\vec{x})$$

where s_p is required to satisfy the boundary condition. In discrete form, the gradient and scalar distribution due to N elements are given by

$$\vec{g} = \nabla s(\vec{x}, t) = \sum_{j=1}^N \frac{\vec{g}_j(\vec{x}_j, t) \delta A_j}{\pi \delta_j^2} f_\delta \left(\frac{\vec{x} - \vec{x}_j}{\delta_j} \right) \quad (17)$$

and

$$s = \sum_{j=1}^N \frac{\delta A_j}{\pi \delta_j^2} \int_V \vec{g}_j(\vec{x}_j, t) \cdot \nabla G(\vec{x} - \vec{x}_j) f_\delta \left(\frac{\vec{x} - \vec{x}_j}{\delta_j} \right) d\vec{x} \quad (18)$$

To evolve the element gradient, we take the gradient of equation (14):

$$\frac{d\vec{g}}{dt} = -\vec{g} \cdot \nabla \vec{u} - \vec{g} \times \vec{\omega} + \alpha \nabla^2 \vec{g} \quad (19)$$

Attempts to integrate equation (19) directly proved problematic [10,12]. A successful alternative is the conservative approach, in which the magnitude and direction of $\vec{g}$ are transported separately. Expressing the local gradient as $\vec{g}_j = |\vec{g}_j| \vec{n}$, where $\vec{n}$ is the unit vector in the gradient direction, and defining a material line by the unit vector $\vec{l}$, such that

$\vec{l} \cdot \vec{n} = 0$, we can update the gradient vector by updating its magnitude and direction as follows. For compressible flow, the equation governing the convective change in the magnitude of the gradient and the product of the density and the length of the material element normal to it are identical, showing that:

$$\frac{|\vec{g}|}{|\rho \delta \vec{l}|} = \frac{\delta s_o}{\rho_o \delta A_o}, \text{ where the subscript } o \text{ denotes } t=0, \text{ and } \delta s_o \text{ is}$$

the initial scalar differential across the element. The evolution of the material unit vector is computed by transporting two material points aligned along it. Thus, for an element j , the scalar gradient is given by:

$$\vec{g}(\vec{x}_j, t) = \frac{\delta s_o}{\rho_o \delta A_o} |\rho \delta \vec{l}| \vec{n}_j \quad (20)$$

Diffusion of scalar gradients is modeled by core expansion, similar to the vortex elements.

The gradient of element j may be expressed using its components in the axial, g_z and radial g_r directions. Following a similar approach to that used to obtain the streamfunction.

4. EXAMPLE:

The parameters for the case described next are:

- (1) the Froude number, $F_r = W_o^2 / (gD) = 0.01$, where W_o is axial velocity at pool surface,
- (2) the buoyancy-based Reynolds number is 700,
- (3) the mass stoichiometric ratio, $\phi = 4.0$,
- (4) the heat of reaction normalized by the initial temperature and specific heat, $Q = 70$.

The initial and boundary conditions are:

- (1) the fuel-air density and temperature ratios are: $\rho_f / \rho_a = 0.1975$, $T_f / T_a = 3.0$,
- (2) on the fuel side: $Y_f = 0.9$, $Y_o = 0.0$, $Y_p = 0.1$, and $Y_d = 0.0$,
- (3) on the air side: $Y_f = 0.0$, $Y_o = 0.2$, $Y_p = 0.0$, and $Y_d = 0.8$.

The molar masses are taken to correspond to methane-air mixture: $M_f = 16$, $M_o = 32$, $M_d = 28$, and $M_p = 26.62$. The heat of reaction is reduced from its adiabatic value to account for incomplete combustion and radiative losses, and the fuel temperature at the pool surface is taken to account for intense radiative heating of the evaporated methane.

The interface between the fuel and air is represented using 9 layers of elements. The average length of an element along a layer is 0.02 (nondimensionalized with respect to the pool diameter). The unsteady calculation proceeded with a time step of $\Delta t = 0.02$.

IV.1. The Early Stages and the Mushroom

During the transient stage between the onset of the simulation when the interface is horizontal and the time the flow reaches a stationary state, the interface between the fuel

and air, i.e. the flame, rises to form the familiar mushroom cloud observed in buoyant flows. Although the rising horizontal interface is, in principle, Rayleigh-Taylor unstable, no signs of instability growth is observed. On the other hand, the vertical surface of the plume, where the density gradient is perpendicular to gravity, is where all the vorticity is generated. The vertical vortex sheet, or sleeve since the flow is axisymmetric, rolls up close to the leading edge of the plume. Thus, the underlying flow leading to the formation of this mushroom is a strong vortex ring structure which forms at the leading edge of the rising plume. The ring forms as the vorticity generated along the rising interface spirals around a common center.

Vorticity is generated by the action of the gravity acting on the rising vertical interface [3]. Vorticity generation is best explained using equation (4). In the results presented next, the baroclinic term, corresponding to vorticity generation due to interaction between the density gradient and the hydrodynamics pressure gradient, was not included in the calculation so as to investigate the impact of the interaction between the density gradient and the gravity term. The former is expected to play a secondary role of the gravity term, and we plan to investigate its impact in a future study.

The formation of the spiral appears to be due to the growth of an intrinsic instability along the vertical vortex layer of the Kelvin-Helmholtz type. There is another form of instability in this flow; it is the Rayleigh-Taylor instability of the initial horizontal interface between the fuel and air. As mentioned before, the mushroom appears to be a consequence of the action of the Kelvin-Helmholtz instabilities acting on the vertical vortex sheet. As the mushroom rises above the pool a distance of several pool diameters, the vertical interface rolls up again to form new vortex rings near the pool. Eventually, the effect of the initial mushroom on the flow near the pool surface becomes negligibly weak (in the computations, as the entire mushroom passes through the exit boundary of the domain, which is located 5-6 plume diameters downstream, it is deleted). This starts the stationary state phase of the simulations.

As a time saving tool, we limit the vertical extent of the computational domain and delete the computational elements as they pass through this virtual boundary. The location of this boundary is chosen such that it exceeds the expected average flame height. The latter is estimated from experimental data which relates the normalized flame height to the quantities

Q^* defined as: $Q^* = \frac{\dot{Q}}{\rho_{\infty} c_p T_{\infty} U D^2}$ which characterizes

the overall Richardson number of this flow [16].

IV.2. Time Dependent Structure of the Plume

The flow eventually turns quasi-periodic with the regular formation of large vortical structures forming near the pool. The size of the vortical structures is of the order of the

pool diameter. The frequency of formation of the vortical structures, known as the puffing frequency, has been investigated both experimentally [1, 2] and numerically [3,4]. The dependence of the puffing frequency, f , represented in the form of a Strouhal Number, $S_t = fD/W_o$, on the Froude number has been shown to be reasonably described by $S_t = Fr^{-m}$, with $m = 0.38$ and 0.57 for non-reacting plumes and fire plumes, respectively [1]. Within the computational domain, at any moment in time, there are three or more large structures. While it is a rarely observed event, structures of different sizes and frequency are shed near the pool. Also, pairing between neighboring structures is observed, although infrequently.

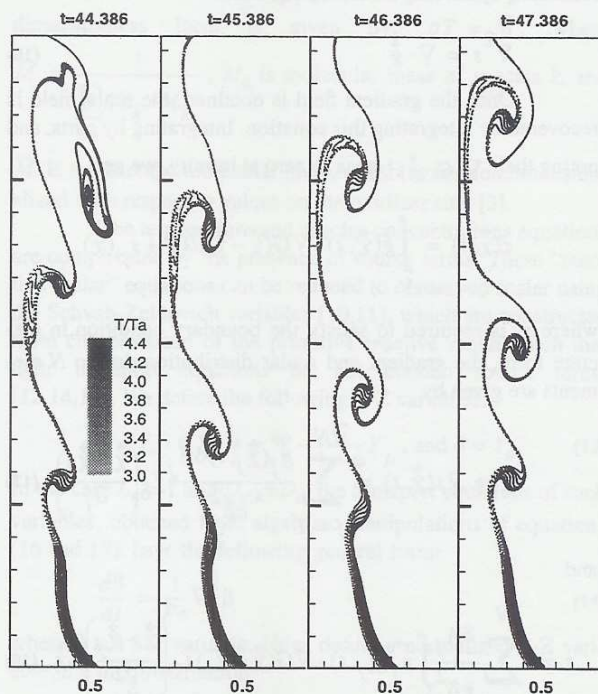


Figure 2. The temperature of the computational elements initially aligned with contours of maximum density gradient.

The evolution of the layers used in the simulations is shown in figure 2, where a color code is also used to indicate the local temperature. All the layers of computational elements are plotted in this figure. The figure shows both the linear and nonlinear stages of the growth of the instability mentioned before, the former is characterized by a downstream propagating wave with a growing amplitude, and the second starts as soon as the wave breaks and a spiral center is formed. The group velocity of the wave is non zero, and it appears to increase as the structure matures. The latter is due to the increase in the buoyancy as lighter products are formed inside these large structures. The figure confirms that the formation of the large structures is due to an instability which grows along

the vorticity layer established by the plume buoyancy. The buoyancy is due in part to the initial buoyancy of the fuel plume and in part to the formation of hot products as the air and fuel mix and burn across the flame.

This figure shows the enormous stretch induced by the rollup of the vorticity layer which is experienced at locations between the large scales. The material initially contained between these layers is now pulled towards the rolling vortical structures. It also shows the combustion enhancement within the large structures where higher temperatures are indicated. The large structures act as intense mixing zones within which the air they entrain is mixed with the fuel, and with fast chemistry, products are formed. Another interesting observation is the relative position of the flame with respect to the vorticity layer. As expected from stoichiometry, the flame is located on the air side of the vorticity layer since it needs more air to consume a certain amount of fuel. Similar observations are reported in reacting jet and shear layer experiments.

The results show that the vortical structures are responsible for air entrainment deep inside the rising fuel layer, leading to high product concentration within these structures. This is observed in figure 3 where the product concentration is shown to reach their maximum values around the outer edges of the evolving structures. These results indicate that while the

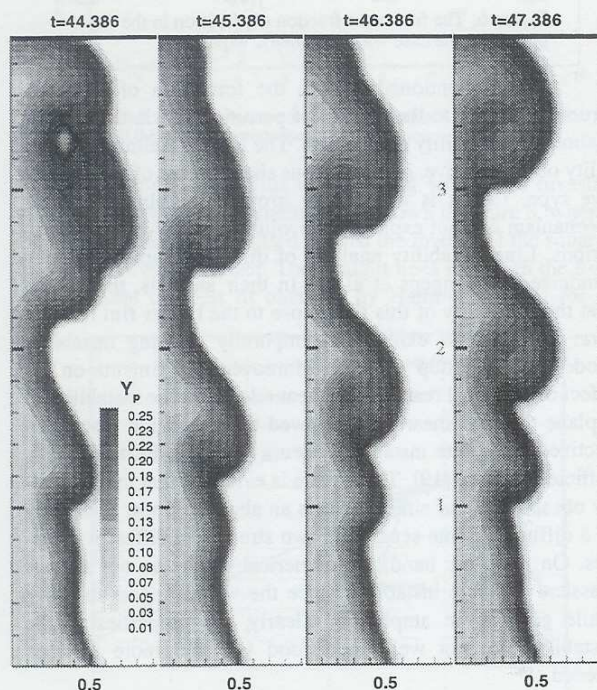


Figure 3. The product mass fraction distribution in the fire plume for the case with volumetric expansion.

products accumulate within the diffusion flame structure, the diffusion flame established between the rising fuel and air is

convoluted by the evolving large vortical structures. The spiraling of the diffusion flame inside the structures is limited by the combined effects of volumetric expansion and vorticity generation, as observed in reacting shear layer results [12]. Contrary to the latter, however, no pairing is observed, most likely due to the downstream (upward) acceleration of the eddies. Product concentration continues to increase inside the eddies until all the fuel is burnt.

The fact that the highest products concentration corresponds to regions of high vorticity is confirmed in figure 4, where vorticity distribution is presented for the same cycle.

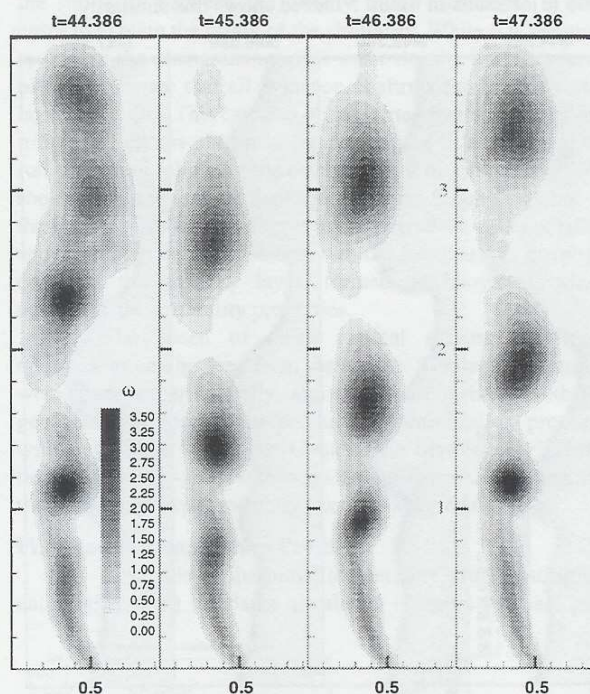


Figure 4. The vorticity distribution in the fire plume for the case with volumetric expansion.

The vorticity distribution in this figure is obtained by differentiating a grid-based velocity field which is recovered from the vortex elements. The establishment of a vertical vortex sheet due to the buoyancy generated torque is clear near the pool, and the first sign of the growing instability is also detectable at this stage. Vorticity continues to accumulate inside these structures as more burning takes place, and the core of maximum vorticity grows. The vortical core vertical acceleration is responsible for its elongation. Eventually, and as most of the fuel is burnt, vorticity generation and its accumulation inside the structures slow down and diffusion lowers the value of the maximum vorticity. Instead, vorticity is spread outwards over the area on the entrained fluid.

The density distribution depicted in figure 5 shows that even before much burning has occurred, i.e., close to the

pool, a sufficiently high rate of vorticity generation should be expected. This is because not only is methane density lower than that of air, methane temperature rising from the pool is assumed to be higher than the ambient temperature due to the implicit effect of the radiative heat flux. This is taken as a boundary condition in this study since it has been shown to be true in many experimental measurements of different pool fires. It should also be mentioned that in our previous studies, we have shown that the similarity between isothermal plumes and fire plumes, over a relatively wide range of conditions stems from the similarity between the density fields in both cases [3]. Results in figure 5 indeed shows this similarity.

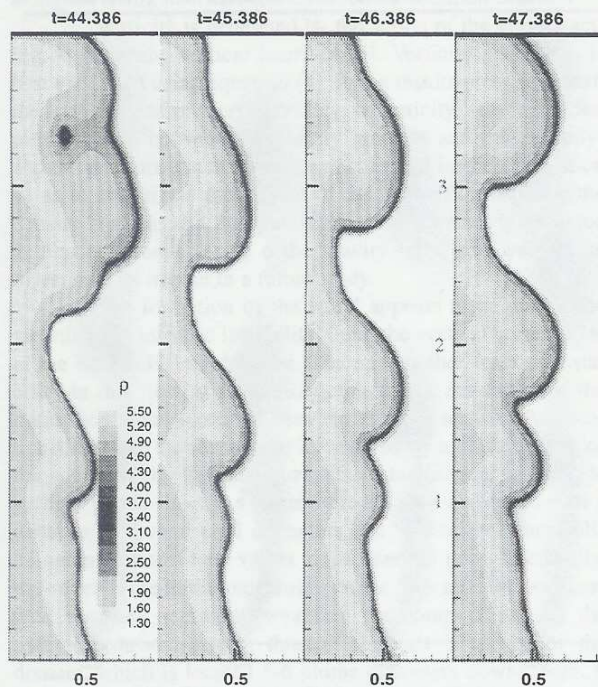


Figure 5. The density distribution in the fire plume for the case with volumetric expansion.

The fuel concentration is depicted in figure 6. Two interesting observations can be made here: (1) fuel concentration can reach zero instantaneously at some locations along the plume axis while its value is nonzero above these locations, and (2) the large structures are responsible for this penetration of the entrained air all the way to the plume axis. The figure also shows that the center of the large eddies is formed of fuel and products and that air entrainment towards these zones is relatively slow. It is worth mentioning here that while there is small fuel zones on the upper sides of the large eddies, there are large fuel zones on the lower side. This is consistent, while not conclusive, with the experimental observation that the higher soot concentration zones should be expected on the lower sides of the large eddies [17].

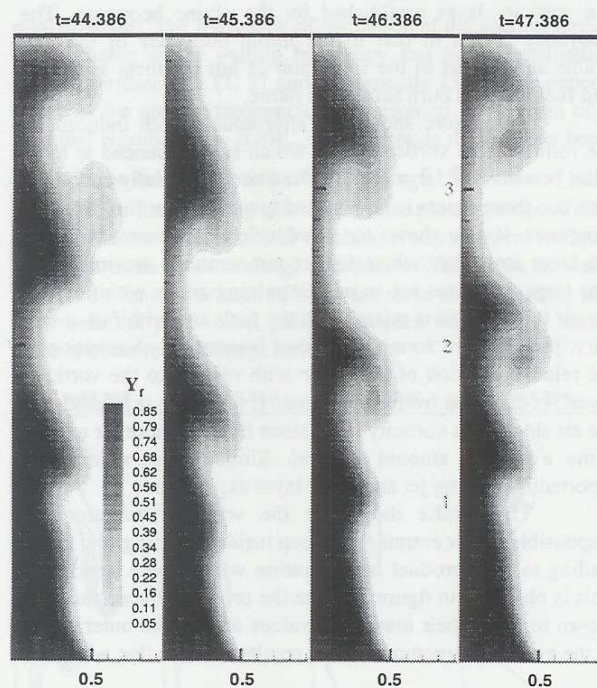


Figure 6. The fuel mass fraction distribution in the fire plume for the case with volumetric expansion.

As mentioned before, the formation of the large structures is due to the growth of perturbation via the Kelvin-Helmholtz instability mechanism. The Kelvin-Helmholtz instability of nonreactive, homogeneous shear flow is of the convective type, i.e., it is a spatially growing instability, and its mechanism can not explain the evolution of self-excited oscillations. Linear stability analysis of diffusion flames has been conducted by Lingens et al.[8]. In their analysis, they found that the instability of this flow close to the burner rim is *absolute*; i.e. the flow exhibits a temporally growing instability mode of zero group velocity. Moreover, experiments on the effect of chemical reaction and heat release on the instability of a plane reactive shear layer showed that transition from convective to absolute instability occurs when the heat release is sufficiently large [19]. Thus, there is evidence that the instability observed in our simulations is an absolute linear instability of a diffusion flame separating two streams of different densities. On the other hand, the numerical results do not show a classical absolute instability since the waves appear to move while growing in amplitude. Clearly, the properties of the instability are not well understood yet, and more work is needed.

IV.3. Puffing Frequency

The above results, presented in terms of the distributions of the computational elements, the chemically active species, and vorticity, show that the most significant feature of the

plume dynamics is the presence of time-dependent structures that determine the instantaneous structure of the fire. The characteristics of the periodic structures, in terms of their size, speed and frequency, observed for the cases investigated in our simulations are supported by experimental studies [20]. Periodic shedding of vortical structures whose size is of the order of magnitude of the pool diameter forming at nearly one pool diameter above the pool surface has been reported in experiments. The Strouhal number of this shedding, based on the buoyancy velocity and pool diameter, corresponding to the shedding frequency of the vortical structures for the case investigated is 2.8. This is obtained from the results shown in figure 7 where the radial velocity component history at a point located at $(0.4D, 0.8D)$ above the pool surface and its Fourier

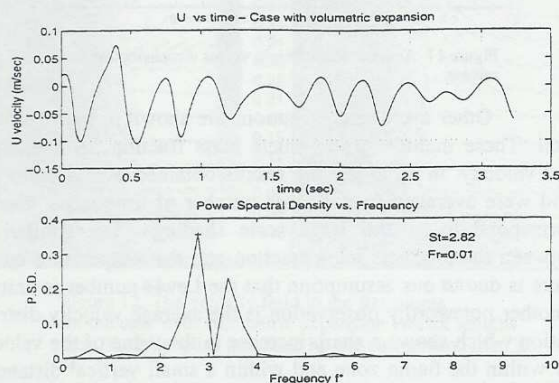


Figure 7. Radial velocity component history at a point located at $(0.4D, 0.8D)$ and the corresponding frequency.

transform are depicted. This result, along with cases investigated in our previous simulations, are shown in figure 8, where the Strouhal number is plotted against the inverse of the square root of the Froude number. The straight lines shown in the figure represent the best fit obtained by Hamins et al.[1] for a

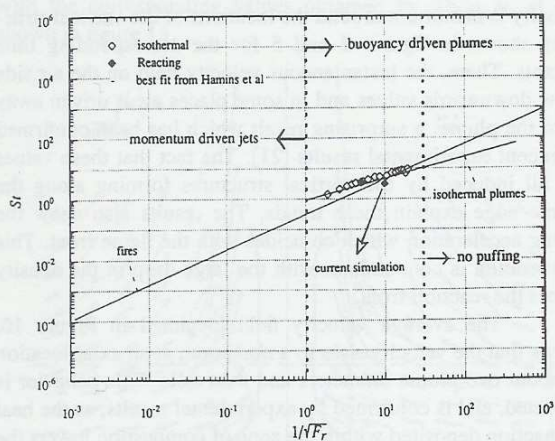


Figure 8. Puffing frequency dependence on the plume diameter for fires and isothermal plume.

large number of experiments.

As mentioned before, the puffing phenomenon is connected with the shedding of large vortical structures as the vorticity layer established by buoyancy rolls up due to a Kelvin-Helmholtz instability. The universal scaling law of this instability suggests that the instability properties are indeed intrinsic to the flow. While it has been hypothesized before that these flows could be linearly unstable, the actual observation of linearly growing waves along the vortex layer has not been reported before. In previous studies, we showed that similar phenomena are observable in isothermal plumes, and we used the similarity between the density fields established in both cases to explain the origin of the similarity. While it is tempting to discuss the phenomena as if it was a density interface instability, we iterate that all evidence confirms that it is a vortex layer instability. This can also be supported by the vorticity distribution contours shown in figure 4 where it is clear that the roll up occurs following the establishment of a finite layer. We should also add that the dependence of the Strouhal number on the Froude number, the latter is a measure of the effect of buoyancy on this flow, is consistent with the fact that it is the properties of the vorticity layer, formed by buoyancy, which determine the instability properties.

The speed of these vortical structures increase upwards, as can be seen from the results. Similar observations were made experimentally, and they are consistent with the generation of more light products as combustion proceeds within the large structures. Comparison between our predictions of the eddy-speed dependence on the vertical location and those obtained experimentally show good agreement.

IV.4. Radial Temperature Profiles

A sample of the radial temperature profiles at different heights along the flame are shown in figure 9. These pro-

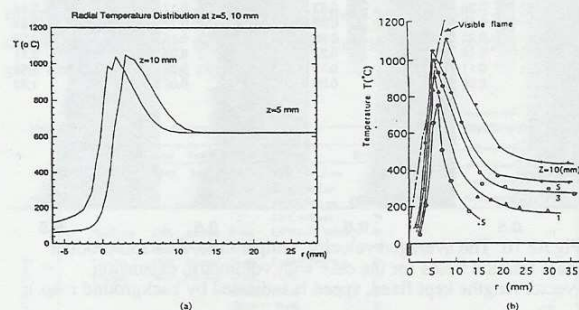


Figure 9. Radial temperature profiles at different heights.

files were obtained for a case for which $T_a = 278$ K and $D = 5.7$ cm. The profiles are reminiscent of diffusion flame temperature profiles in cases when the strain rate exerted on the flame is relatively low. A steep temperature drop is observed on both sides of the reaction front, with higher steepness on the air side. This is in qualitative agreement with corresponding temperature

profiles obtained experimentally by Venkatesh et al. [20], presented in the same plot for a fire of 5.7 cm pool diameter with 1-propanol as fuel. In our case, the fuel-air temperature ratio converges to a value of 3.0 at $r=D/2$, which is the ratio provided to the setup of the fuel-air interface as an initial condition. These profiles serve to show the extent of the flame structure within the fire zone, and to emphasize the significance of the high resolution achieved using the transport element method in capturing the evolving combustion field in the fire.

IV.5 Average Flame Height

The observed flame height has been defined as the point above the pool where most of the fuel has burnt. Our simulations show that the flame height fluctuates around a plane nearly three diameters above the pool, as seen in the distribution of the instantaneous fuel mass fraction shown in figure 6, and the averaged fuel mass fraction distribution presented in figure 10. As mentioned before, flame fluctuations are due to

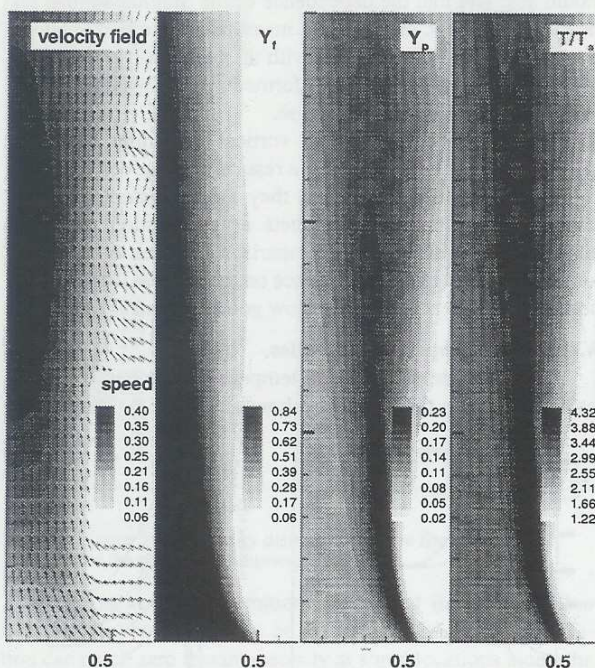


Figure 10. The averaged velocity, temperature, fuel and product mass fraction fields for the case with volumetric expansion. (vector lengths kept fixed, speed is indicated by background map.)

the shedding of the large structures which tend to divide the evaporating fuel plume into segments and at times separate parcels of the plume from the rest. The flame height, Z_{fl}/D , has experimentally been correlated to the normalized heat of reaction

$$Q^* = \frac{\dot{Q}}{\rho_{\infty} c_p T_{\infty} U D}^2 \text{ according to the plot shown in}$$

Figure 11 which was compiled by Zukoski[16]. For our case,

the flame height of $Z_{fl}/D=3$, corresponding to $Q_D=1.0$, is in good agreement with the experimental data.

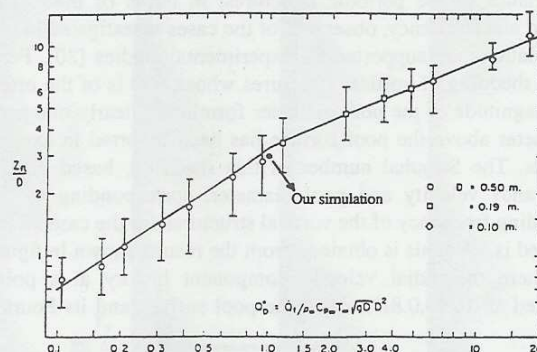


Figure 11. Average flame height versus dimensionless heat release.

Other average distributions are shown in figure 10 as well. These include the products mass fraction, temperature and velocity. In all cases, the results obtained over a uniform grid were averaged over a large number of time steps which correspond to several large scale shadings. The similarity between the products mass fraction and the temperature contours is due to our assumption that the Lewis number is unity. Another noteworthy observation is the average velocity distribution which shows a sharp increase in the value of the velocity within the flame zone and within a small vertical distance above the pool.

IV.6 Velocity Field

The instantaneous velocity fields at dimensionless times $t=45.386$ and $t=47.386$ are shown in figure 12 to help shed some light on the properties of the entrainment field. The velocity on the air side indicates the entrainment of air by the vortical structures into the reacting zone. The entrainment velocity is maximum around the locations of the vortical structures shown in figures 2 and 5 for the corresponding time instants. There, the instantaneous velocity field on the air side show downwards values and in some places air is driven away from the plume; a surprising result which has been confirmed by recent experimental results [21]. The fact that these values are all induced by the vortical structures forming along the plume edge explain these trends. The results also show the strong acceleration which coincides with the flame front. This acceleration is concomitant with the large drop in the density across the reaction front.

The average velocity field, depicted in Figure 10, shows that the velocity rises to a maximum at an axial location of about two plume diameters and then falls. This behavior is expected, and is confirmed by experimental results, as the heat of reaction deposited within the zone of combustion lowers the density of the products and accelerates the flow, followed by a zone of nearly pure entrainment without measurable combus-

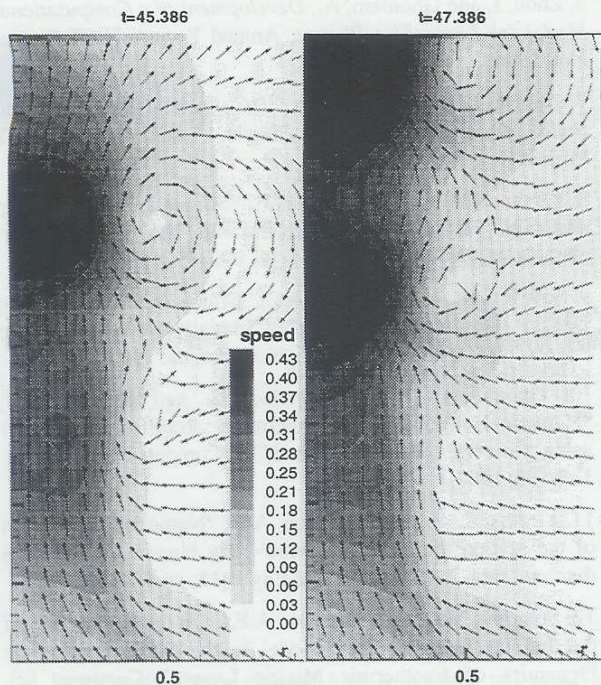


Figure 12. The velocity field in the fire plume for the case with volumetric expansion (vector lengths kept fixed, speed is indicated by the background map.).

tion which leads to lower temperatures and lower velocity. The ratio of the velocity of the air entrained into the flame zone at an axial location of $0.7D$ is about four times the velocity of the air entrained at the pool base. This agrees with corresponding values obtained by Zhou et al. [21] for a 7.1 cm toluene fire. Furthermore, the values of the velocity corresponding to a 7.1 cm diameter pool ranges from ~ 6 cm/s on the air side at the pool base to 25 cm/s at $z \sim 0.7D$, which is in good agreement with the corresponding values obtained by Zhou et. al and shown in figure 13.

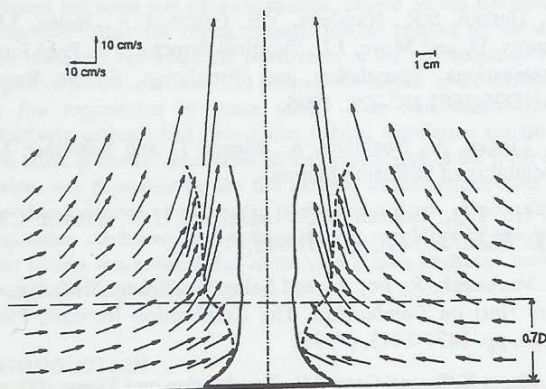


Figure 13. Velocity field at pool base of a 7.1 cm toluene fire obtained from ZHOU, X.C and GORE, J.P., Combustion and Flame 100:52-60(1995)

IV.7. Entrainment

Finally we use the average velocity field to compute the entrainment rate into the fire zone. The entrainment rate is defined as the mass flow rate of air entering the flame region at

a well defined radius; $\dot{m}_{ent} = \frac{d}{dz} \left(\int_0^R 2\pi r w dr \right)$. Clearly the

selection of the radius at which the integration is performed is crucial. Figure 14 shows the entrainment rate computed from our results for different choices of the radius R , normalized by the Froude number, the stoichiometric ratio and the fuel mass flow rate injected at the pool. On the same plane, we show the

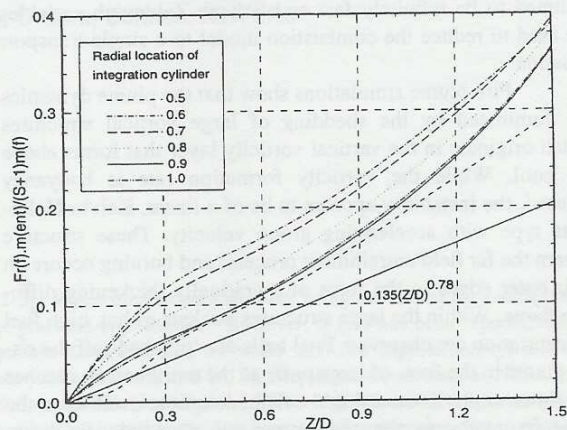


Figure 14. Mass entrained into fire as a function of elevation.

results from Ref. [21]. Plots showing a wide range of entrainment rates have been published in the literature (see [16]), confirming the need for a precise definition of this important quantity. Finally, we show on figure 15 a sample of this wide discrepancy in the experimental results, along with Delichatsios correlation and our results.

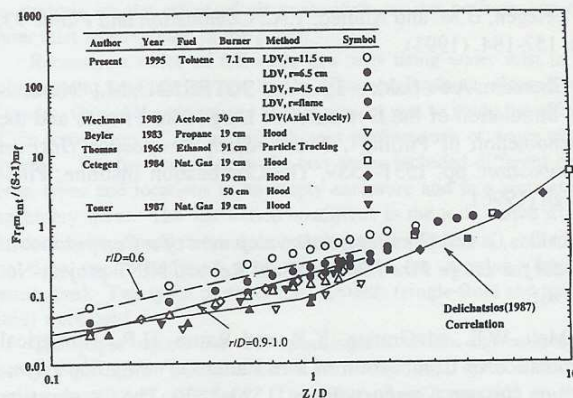


Figure 15. Comparison of present numerical results concerning entrainment mass-flow rates with experimental data and Delichatsios's correlation from the literature.

5. CONCLUSIONS

This paper describes recent progress on the formulation of a Lagrangian computational approach to simulate large fire plumes. The approach is based on the vortex-element method in which the unsteady Navier-Stokes equations are solved in their vorticity-transport form using a set of moving computational elements carrying time-dependent vorticity. Vorticity is generated due to the buoyancy-induced torque according to the evolving density and pressure gradients. The energy and species transport equations are solved using the transport-element method in which, to maintain compatibility with the vortex method, the gradients of the scalars are evolved by convection and diffusion. The fuel oxidation reaction is assumed to be infinitely fast and Schvab-Zeldovich variables are used to reduce the combustion model to a single transport equation.

Fire plume simulations show that the plume dynamics are dominated by the shedding of large vortical structures which originate in the vertical vorticity layer that forms above the pool. While the vorticity formation rate is buoyancy induced, the instability appears to be of a linear, Kelvin-Helmholtz type with accelerating group velocity. These structures govern the far field entrainment process, and burning occurs on their outer edges in the form of a gradually thickening diffusion flame. Within the large structures pockets of hot, high fuel concentration are observed. Fuel balls are 'pinched' off the rising plume in the form of large puffs as the entrained air reaches the plume centerline causing the flame height to oscillate at the same frequency as the shed structures. Dynamic similarity between isothermal and intense fire plume is discussed in light of the density fields in both cases. Predictions of the puffing frequency, averaging flame height and oscillations, and motion of the large structures compare well with experimental data.

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